
Beyond the Cradle: Multi-Objective Optimization and System Synergies of a Self-Sustaining Space Settlement

Summary

Humanity's expansion into a multi-world species requires the establishment of permanent extraterrestrial settlements capable of supporting tens of thousands of inhabitants. This study addresses the ambitious challenge of building a 100,000-person lunar colony by 2050, which entails the transport of 100 million metric tons of materials and the deployment of life-support systems, all while adhering to strict cost, timeline, and environmental constraints. To guide the decision-making process, we develop an integrated optimization framework that balances fiscal responsibility, timely delivery, and planetary stewardship for interplanetary colonization.

The framework incorporates: **The Independent Logistics Channel Model**, which evaluates Space Elevator and rocket systems as parallel transport channels and optimizes mass flow across cost, timeline, and emissions objectives. Cost projections account for exponential decay in rocket pricing (forecasted to reach \$1,606 per kg by 2050) and the Levelized Cost of Transport for Space Elevators, which decreases from \$13,200 to \$7,450 per tonne due to technological maturation. The **Phased Dynamic Evolution Framework** models long-term water sustainability by simulating population growth, escalating per capita water demand from 20 to 250 liters per day, and improving recycling efficiency from 85% to 99.7% over the course of seven decades.

Using **Differential Evolution**, we identify **Pareto-optimal strategies** that balance the trade-offs between competing objectives. The hybrid approach, which combines 537,000 tonnes annually through the Space Elevator and 439,200 tonnes via rockets, completes the colony in 102 years at a total cost of \$38.9 trillion, reducing costs by 55%. **Stochastic reliability analysis** confirms the robustness of the strategy, with only 2.4 years of delays and a 0.8% increase in cost under non-ideal conditions. **Water demand projections** reveal a critical 24-year stress window, during which demand peaks at 826,500 tonnes, necessitating rocket supplementation. Environmental modeling shows that prioritizing emissions at 26.2% of the total objective eliminates rocket usage but extends the timeline by 29%.

In conclusion, this hybrid strategy, combined with phased water management, provides the optimal path for sustainable lunar colonization, effectively balancing economic, technical, and environmental goals. As humanity ventures beyond the cradle, the principles explored in this study will serve as the foundation for designing self-sustaining space settlements on the Moon and beyond. In the future, further refinement of these models, particularly with respect to evolving technological capabilities and geopolitical conditions, will be essential in ensuring the long-term viability of interplanetary settlements. This work contributes to the broader vision of creating resilient, efficient, and sustainable colonies, paving the way for humanity's expansion into the cosmos.

Keywords: Lunar colonization; Space Elevator; Multi-objective optimization; Differential Evolution; Water sustainability; Environmental impact modeling

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1 Introduction

1.1 Background: Logistics at the Planetary Scale

Establishing a permanent Moon Colony marks a transition from exploration to sustained civilization. Yet the mission specifications indicate that supporting a population of 100,000 by 2050 requires transporting on the order of 10^8 metric tons of infrastructure and supplies, a scale that elevates lunar logistics into a planetary-class systems engineering challenge.

This challenge is fundamentally defined by the architectural dichotomy illustrated in Figure 1. Traditional chemical propulsion relies on high-energy Trans-Lunar Injection Trajectories, where the physics of the Vis-viva equation dictates exponential propellant penalties for payload delivery. In contrast, the Space Elevator system replaces these discrete ballistic arcs with a permanent mechanical infrastructure. Spanning 100,000 km from the Earth Port to the Galactic Harbour (Apex Anchor), this structure utilizes a GEO Transfer Node to balance dynamics and facilitate continuous payload ascent. While this tether-based approach bypasses the tyranny of the rocket equation, it introduces new complexities in large-scale deployment and material durability.

Therefore, lunar logistics is not a binary technology choice but an allocation problem. A viable strategy must balance the flexible, rapid deployment of rockets with the high-throughput, propellant-efficient capacity of the elevator, creating a robust supply chain from LEO to the Lunar Base. This hybrid perspective reframes the challenge as an optimization of mass allocation under cost and time constraints. Accordingly, we formulate a quantitative model to determine the optimal transport mix across development phases. The resulting solution provides a scalable pathway toward sustaining a 100,000-person lunar settlement.

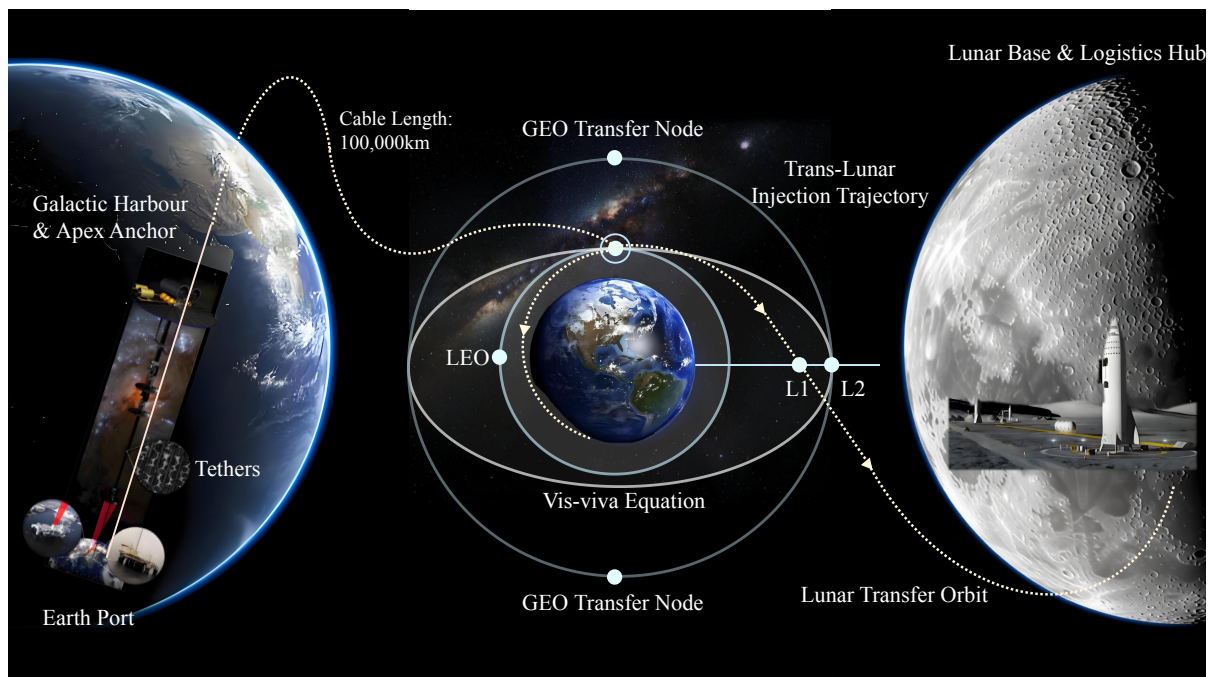


Figure 1: Conceptual Architecture of the Hybrid Earth-Lunar Logistics Network.

1.2 Restatement of the Problem

To provide the MCM Agency with a decision-ready lunar logistics roadmap, we restate the mission as four distinct modeling tasks:

- **Task 1: Strategic Scenario Benchmarking**

Problem: Quantify and compare total cost and project duration for delivering 10^8 metric tons under three scenarios: Rocket-only, Elevator-only, and Hybrid.

⇒ **Solution:** Develop the **Independent Logistics Channel Model (ILCM)**. This multi-objective framework integrates distinct cost structures to determine the optimal flow allocation ratio and economic break-even points.

- **Task 2: Operational Resilience Under Non-Ideal Conditions**

Problem: Assess how the preferred strategy changes under operational imperfections and quantify sensitivity to reliability degradation.

⇒ **Solution:** Perform a **Reliability Sensitivity Analysis**. By introducing stochastic parameters, we map the failure topology to identify thresholds where the strategy must shift.

- **Task 3: Water Sustainability After Occupation**

Problem: Estimate the water requirement for a 100,000-person colony and model the transition from Earth-dependent resupply to ISRU-enabled self-sufficiency.

⇒ **Solution:** Construct a **Technological Saturation Model** based on logistic growth. We compute net external demand to ensure zero-deficit operations.

- **Task 4: Planetary Environmental Impact Assessment**

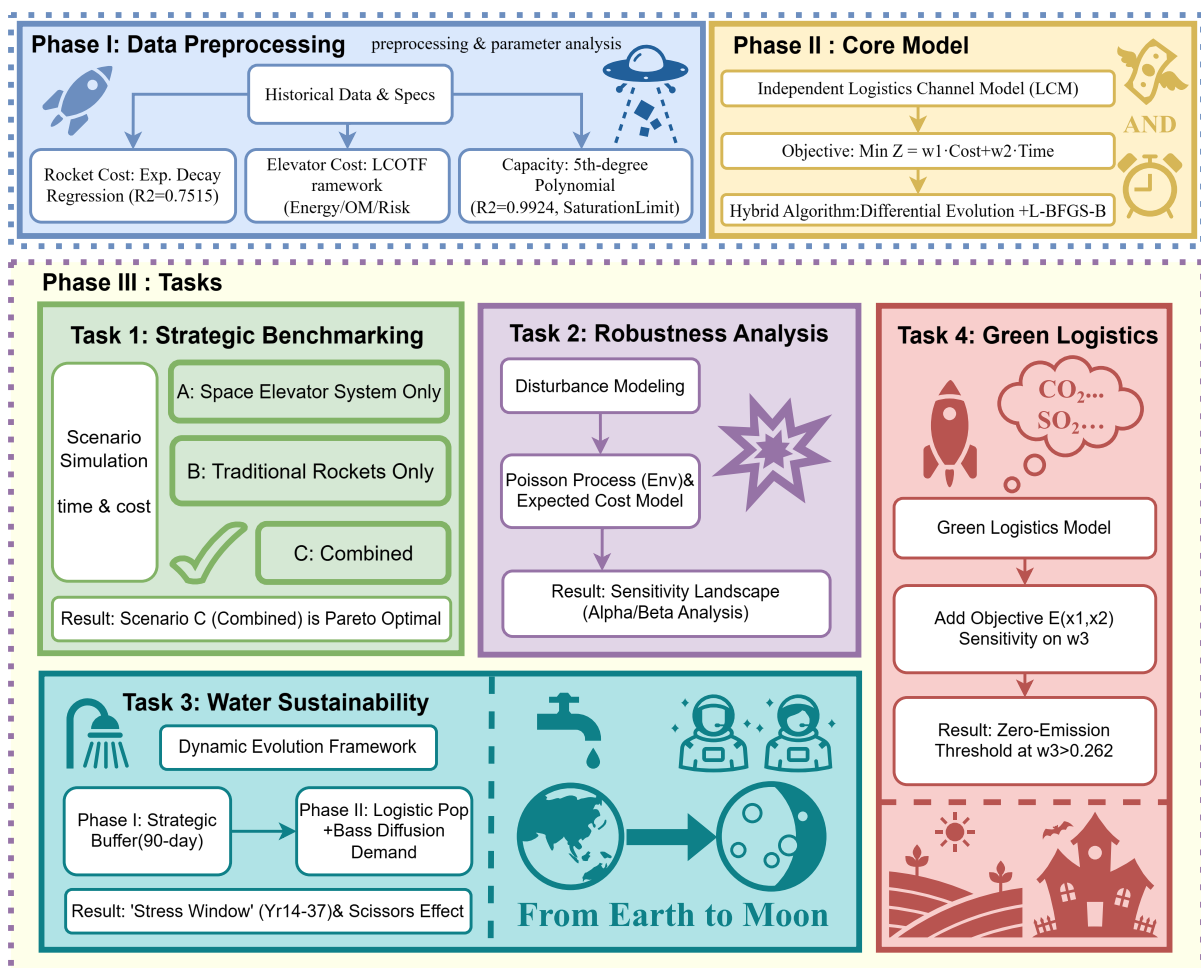


Figure 2: Overall Workflow of the Lunar Colony Optimization Framework

Problem: Evaluate the cumulative environmental footprint on Earth (e.g., atmospheric emissions, launch debris) for achieving the colony. We must discuss how to adjust the logistics model to minimize ecological damage.

⇒ *Solution:* Integrate an **Environmental Impact Module** based on Lifecycle Assessment (LCA). We adjust the original objective function to penalize high-emission transport modes, proposing a "Green Logistics" optimization strategy.

Figure 2 illustrates the approach used to optimize lunar colony logistics and sustainability. Phase 1 preprocesses data to assess transport costs and capacity. In Phase 2, the Independent Logistics Channel Model, combined with Differential Evolution, optimizes mass flow while balancing cost, timeline, and emissions. Phase 3 evaluates transport strategies, system robustness, water sustainability, and environmental impact. Each phase builds on the previous to form a comprehensive framework for planning a sustainable lunar colony.

2 Assumptions

To establish a solvable baseline model for the interplanetary logistics system, we make the following rational assumptions.

- **Assumption 1: Independent and Capacity-Constrained Channels.**

We treat the throughputs of Traditional Rockets (x_1) and the Space Elevator (x_2) as independent decision variables, each constrained by its own infrastructure limit.

Justification: Since the two systems rely on different resources (launch pads vs. tethers), they do not interfere, allowing x_1 and x_2 to vary independently within their physical constraints (launch frequency and tether capacity).

- **Assumption 2: Ideal Operational State.**

We assume ideal operations and ignore stochastic disruptions (e.g., weather, mechanical failures) in the baseline model.

Justification: This simplifies the model and provides an upper-bound benchmark for minimum time and cost, while uncertainty is addressed later in the Robustness Analysis section.

- **Assumption 3: Volume-Independent Unit Costs.**

Unit costs ($C_1(t)$, $C_2(t)$) are assumed to be volume-independent.

Justification: Ignoring marginal cost effects (e.g., congestion) maintains the convexity of the objective function, ensuring a global optimum.

- **Assumption 4: Launch Frequency Saturation.**

Global rocket launch frequency is assumed to saturate at a stable upper limit ($L_{\max}$) after 2050.

Justification: Unlimited growth is unrealistic; this assumption reflects the finite capacity of Earth-based launch infrastructure and avoids over-extrapolation.

3 Model Preparation and Detailed Parameter Analysis

To ensure the model's reliability, we analyzed four key parameters: unit transportation costs ($C_1(t)$, $C_2(t)$) and annual capacities (x_1 , x_2). The estimation logic and data sources are outlined below, with symbols and notations summarized in Table 1.

Table 1: Notations used in this paper

Symbol	Description	Units
x_1	Annual mass transported via Traditional Rockets	ton/year
x_2	Annual mass transported via Space Elevator	ton/year
M_{goal}	Total target mass to be transported (Fixed value)	ton
$C_1(t)$	Unit transport cost of Traditional Rockets	\$/ton
$C_2(t)$	Unit transport cost of Space Elevator (including tugs)	\$/ton
N_{ports}	Number of Galactic Harbours	-
C_{single}	Annual capacity limit of a single Galactic Harbour	ton/year
L_{max}	Maximum annual capacity of global rocket launch sites	ton/year
T_{max}	Normalized time benchmark (Duration using only Elevators)	Year
T_{min}	Minimum project duration (using maximum combined capacity)	Year
w_1, w_2	Weighting coefficients for cost and time ($w_1 + w_2 = 1$)	-

* *There are some variables that are not listed here and will be discussed in detail in each section.

3.1 Unit Cost of Traditional Rocket Transport $C_1(t)$

The unit cost of rocket launches has witnessed a paradigm shift over the past six decades. By analyzing historical Low Earth Orbit cost data [1], we observed a trajectory starting with the Saturn V era at approximately \$18,500 per kilogram and the Space Shuttle era at \$54,500 per kilogram, before dropping significantly to the modern SpaceX Falcon architecture at roughly \$1,400 per kilogram. Recognizing this distinct downward trend driven by reusability, we applied an exponential decay regression model to characterize the cost reduction and project launch costs for the year 2050.

Adopting an exponential decay framework with the timeline normalized to the year 2000, we performed ordinary least squares regression on log-transformed historical data. The derived functional relationship and governing coefficients are synthesized as follows:

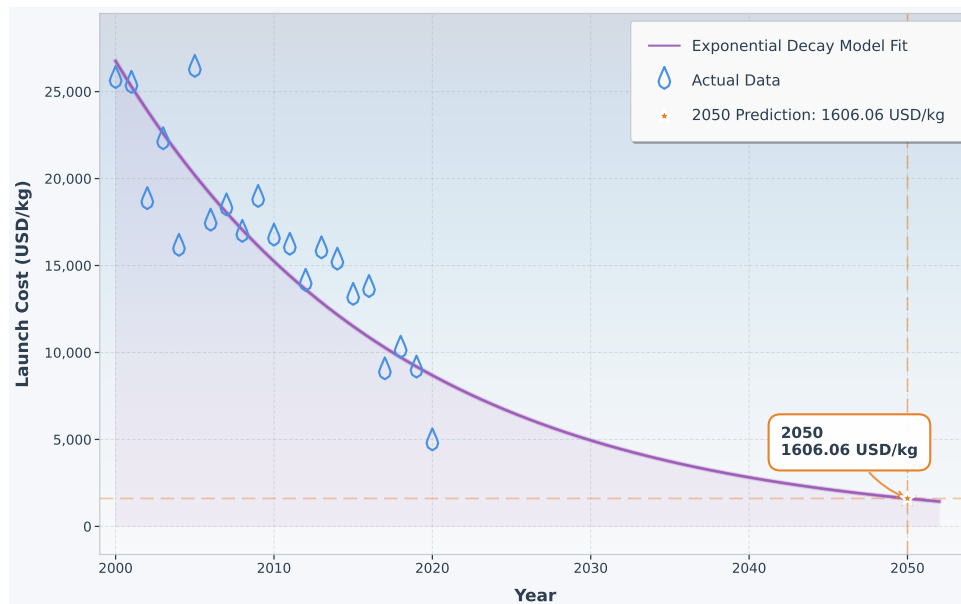


Figure 3: Exponential decay regression of rocket launch costs ($C_1(t)$)

$$\begin{cases} C_1(t) = a \cdot e^{-b(t-2000)} \\ a = 26,754.89 \text{ USD/kg} \\ b = 0.0563 \text{ year}^{-1} \end{cases} \quad (1)$$

Substituting parameters yields the final equation ($R^2 = 0.7515$), confirming technological scaling and reusability as key cost drivers. Extrapolating to 2050, the projected LEO transport cost is:

$$C_1(2050) = 26,754.89 \cdot e^{-0.0563 \times 50} \approx 1,606 \text{ USD/kg}$$

This projection marks a 99.4% drop from Shuttle-era peaks**, reflecting the industry shift toward affordable space access. Historical trends, the fitted model, and the 2050 forecast are illustrated in Figure 3.

3.2 Unit Cost of Space Elevator Transport $C_2(t)$

Unlike propellant-limited rocketry, the Space Elevator relies on beamed-energy climbers, largely decoupling energy expenditure from payload mass. We adopt a Levelized Cost of Transport (LCOT) framework [2] and decompose the unit cost into four components: energy, O&M, capital amortization, and risk/insurance. The total unit cost is defined as:

$$C_2(t) = C_{\text{energy}}(t) + C_{\text{O\&M}}(t) + C_{\text{cap}}(t) + C_{\text{risk}}(t) \quad (2)$$

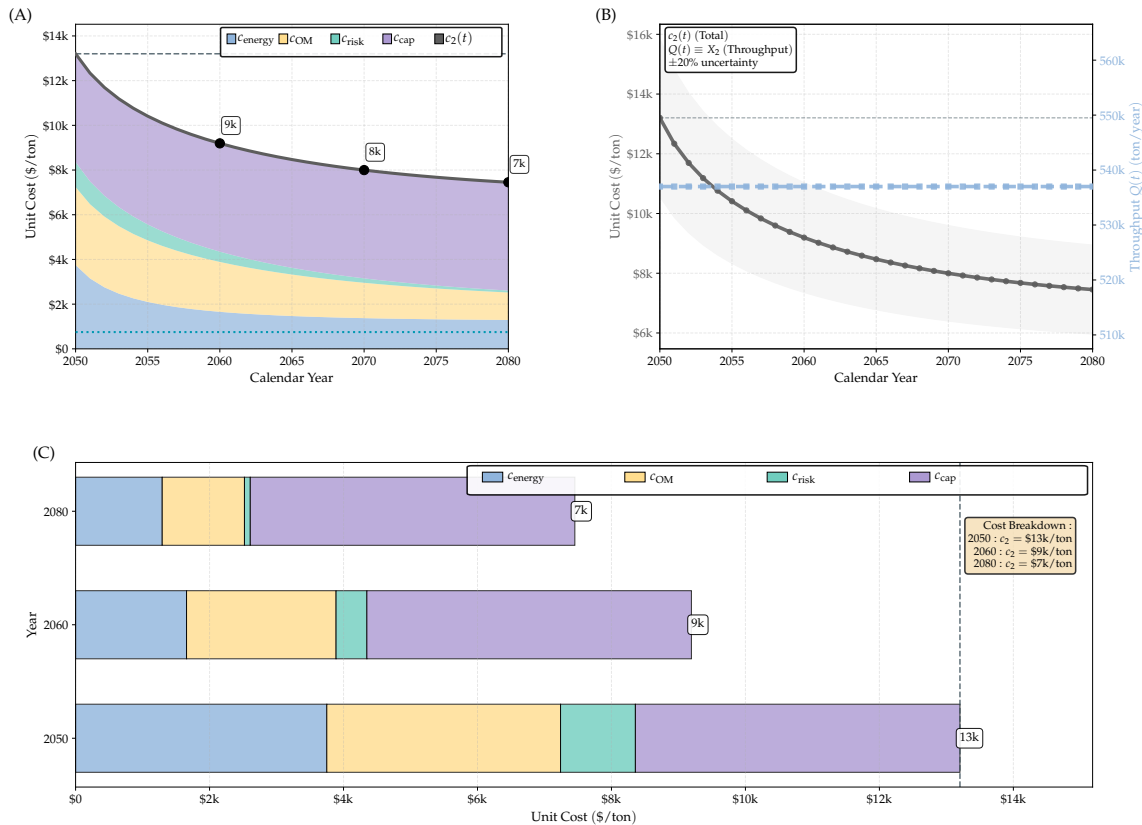


Figure 4: Evolution of Space Elevator Unit Cost ($C_2(t)$). (A) Component breakdown (2050–2080). (B) Cost trajectory with $\pm 20\%$ uncertainty bands. (C) Structural comparison at key milestones (2050, 2060, and 2080).

3.2.1 Baseline operation

In the steady-state baseline, the elevator operates at rated annual throughput:

$$Q(t) \equiv X_2, \quad X_2 = 537,000 \text{ tonnes/year.} \quad (3)$$

Hence, temporal variation in $C_2(t)$ arises from efficiency maturation, learning effects, and reliability growth, rather than throughput scaling.

(1) Energy cost [$C_{\text{energy}}(t)$] Driven by the gravitational potential energy to geostationary orbit, this component is modulated by the total system efficiency $\eta_{\text{tot}}(t)$ and electricity pricing [3]:

$$C_{\text{energy}}(t) = \frac{e_{\text{spec}}}{\eta_{\text{tot}}(t)} p_e \cdot 1000. \quad (4)$$

We set $e_{\text{spec}} = 15 \text{ kWh/kg}$ and $p_e = 0.05 \text{ USD/kWh}$. Efficiency matures as:

$$\eta_{\text{tot}}(t) = \eta_0 + (\eta_{\infty} - \eta_0) (1 - e^{-k_{\eta} t})$$

$$(\eta_0 = 0.20, \eta_{\infty} = 0.60, k_{\eta} = 0.10)$$

(2) O&M cost [$C_{\text{O\&M}}(t)$] This component encompasses fixed infrastructure upkeep and variable handling expenses:

$$C_{\text{O\&M}}(t) = \frac{F_{\text{O\&M}}(t)}{X_2} + v_{\text{O\&M}}(t). \quad (5)$$

Learning-by-doing [4] reduces both terms:

$$F_{\text{O\&M}}(t) = F_{\infty} + (F_0 - F_{\infty})e^{-k_{\text{learn}} t},$$

$$(F_0 = 8.0 \times 10^8, F_{\infty} = 4.0 \times 10^8, k_{\text{learn}} = 0.10),$$

$$v_{\text{O\&M}}(t) = 2000 e^{-0.05t} \text{ USD/tonne.}$$

With $Q(t) \equiv X_2$, the decline in per-unit O&M is driven by learning effects rather than throughput growth.

(3) Capital amortization [$C_{\text{cap}}(t)$] Calculated using the Capital Recovery Factor (CRF) [5]:

$$C_{\text{cap}}(t) = \frac{I \cdot \text{CRF}(i, n)}{X_2} = \frac{I \cdot i(1+i)^n}{[(1+i)^n - 1] X_2}. \quad (6)$$

We use $I = 4.0 \times 10^{10} \text{ USD}$, $i = 0.05$, and $n = 30 \text{ years}$.

(4) Risk & insurance [$C_{\text{risk}}(t)$] Aggregates insurance premiums and probabilistic failure costs, modeled via a reliability growth curve [6]:

$$C_{\text{risk}}(t) = \frac{\Pi(t)}{X_2} + \frac{\lambda(t)L_{\text{loss}}}{X_2}. \quad (7)$$

Reliability improves as $\lambda(t) = \lambda_0 e^{-m_{\text{rel}} t}$ with $\lambda_0 = 0.01$ and $m_{\text{rel}} = 0.15$. Insurance premiums decay with maturity:

$$\Pi(t) = 5.0 \times 10^8 e^{-0.08t} \text{ USD/year}, \quad L_{\text{loss}} = 1.0 \times 10^{10} \text{ USD.}$$

3.2.2 Integrated form

Substituting Eqs. (4)–(7) into Eq. (2) yields the complete LCOT expression:

$$C_2(t) = \frac{e_{\text{spec}}}{\eta_{\text{tot}}(t)} p_e \cdot 1000 + \left[\frac{F_\infty + (F_0 - F_\infty)e^{-k_{\text{learn}}t}}{X_2} + v_{\text{O\&M}}(t) \right] + \frac{I \cdot i(1+i)^n}{[(1+i)^n - 1] X_2} + \frac{\Pi(t) + \lambda_0 e^{-m_{\text{rel}}t} L_{\text{loss}}}{X_2}. \quad (8)$$

The model projects a cost decline from \$13.20/kg (2050) to \$9.19/kg (2060), stabilizing at \$7.45/kg by 2080. This 43.5% reduction is driven by efficiency maturation, technological learning, and reliability growth at rated throughput.

Figure 4 visualizes these dynamics. Panel (A) illustrates the transition from capital dominance to balanced operations driven by learning and reliability gains. Panel (B) quantifies this cost reduction with $\pm 20\%$ uncertainty bands. Finally, Panel (C) confirms the structural shift from a capital-intensive 2050 regime to the efficiency-optimized 2080 benchmark.

3.3 Annual Rocket Transportation Capacity (X_1)

To estimate annual capacity, we model global orbital launch frequency using historical data¹. We restrict our regression analysis to the post-2000 era, which captures the advent of modern commercial spaceflight and better represents contemporary growth compared to the geopolitically constrained pre-2000 period.

We evaluated various regression frameworks, including linear, exponential, logarithmic, and polynomial functions. The fifth-degree polynomial model proved optimal, achieving an R^2 of 0.9924 and an RMSE of 5.85 launches. This confirms the model explains over 99.2% of the variance with minimal uncertainty relative to the data range (55–329 launches). As shown in Figure 5, both fourth- and fifth-degree polynomials significantly outperform other forms ($R^2 > 0.99$), effectively capturing the accelerating growth trend from 2000 to 2025.

The selected polynomial regression model is:

$$N_{\text{launches}}(t) = \sum_{i=0}^5 a_i t^i \quad (9)$$

The unit transportation cost is modeled as a fifth-order polynomial, where t is the year and the coefficients a_i are detailed in Table 2:

Table 2: Coefficients of the Polynomial Regression Model

a_5	a_4	a_3	a_2	a_1	a_0
-2.9573×10^{-5}	3.0368×10^{-1}	-1.2468×10^3	2.5585×10^6	-2.6239×10^9	1.0760×10^{12}

In 2050, the estimated annual launch frequency is $N_{\text{launches}}(2050) = 12,996$ launches. Assuming a payload capacity of 150 tons per launch as specified in the problem, the total annual transportation capacity is calculated as:

¹ <https://planet4589.org/space/stats/out/tabla.txt>

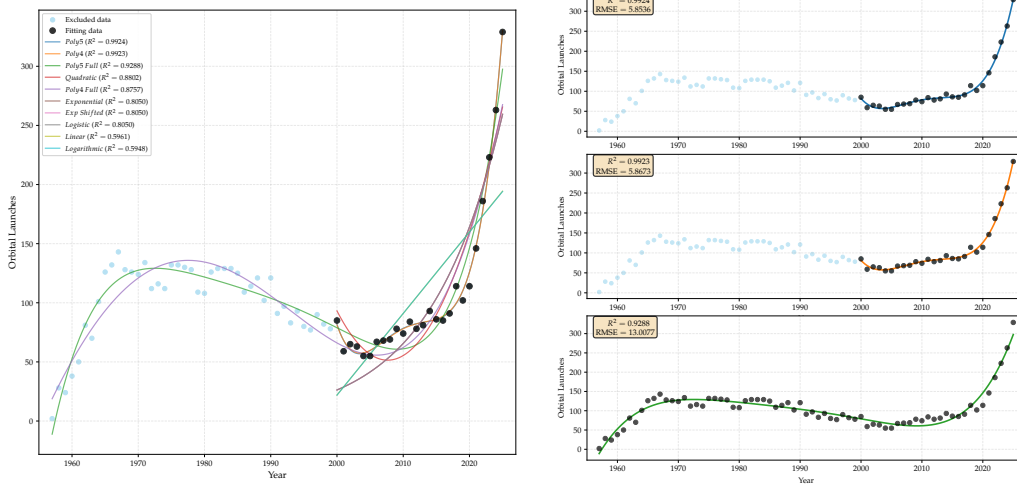


Figure 5: Comparison of regression models for global annual orbital launch frequency

$$Max(X_1) = N_{\text{launches}}(2050) \times M_{\text{payload}} = 12,996 \times 150 = 1.95 \times 10^6 \text{ tons/year} \quad (10)$$

Consistent with Assumption 5, which assumes launch frequency saturates after 2050, we set the computed value as a constant upper bound L_{max} for the rocket logistics channel. Thus, the global capacity is fixed at 1.95×10^6 tons/year throughout the construction phase, representing the limit of a fully mature launch infrastructure.

3.4 Annual Space Elevator Transportation Capacity (X_2)

The problem specifies a nominal throughput of 179,000 metric tons per Galactic Harbour. For the baseline optimization, we treat the Space Elevator capacity as a time-invariant rated value rather than a decision variable. This steady-state assumption models a fully operational system and avoids introducing unsupported expansion controls such as additional climbers. Accordingly, the total annual capacity across three parallel Galactic Harbours is imposed as a constant constraint:

$$X_2 = 3 \times 179,000 = 537,000 \text{ tons/year}, \quad (11)$$

where $0 \leq x_2 \leq X_2$ is enforced throughout the planning horizon.

This comprehensive parameter analysis provides the foundation for our model optimization process. The table is designed to span multiple pages if needed, ensuring readability and accessibility of all parameter information.

4 Task 1: Comparative Analysis of Material Transport Strategies

To transport 10^8 metric tons of materials to the lunar colony, we formulate a static multi-objective optimization model that quantifies cost-time trade-offs under three scenarios: Space Elevator only, traditional rockets only, and a combined approach.

4.1 Construction of the Static Cost-Time Optimization Model

We propose the ILCM, modeling the Space Elevator and rockets as parallel transport channels and minimizing a weighted objective of normalized cost and time. In the baseline, we assume ideal operations and omit stochastic disruptions such as tether sway, launch failures, and mechanical downtime, addressed in sensitivity analysis.

Mathematical Formulation

The ILCM is formulated as a static non-linear optimization problem aiming to minimize the weighted sum of normalized cost and normalized time:

$$\begin{aligned} \min_{x_1, x_2} \quad & Z = w_1 \underbrace{\left(\frac{\int_{2050}^{\frac{M_{\text{goal}}}{x_1+x_2}+2050} c_1(t)x_1 dt + \int_{2050}^{\frac{M_{\text{goal}}}{x_1+x_2}+2050} C_2(t)x_2 dt}{L_{\text{max}} \int_{2050}^{T_{\text{min}}+2050} c_1(t) dt} \right)}_{\mathcal{C}(x_1, x_2)} + w_2 \underbrace{\left(\frac{M_{\text{goal}}}{T_{\text{max}}(x_1 + x_2)} \right)}_{\mathcal{T}(x_1, x_2)} \\ \text{s.t.} \quad & \begin{cases} 0 \leq x_1 \leq L_{\text{max}}, \\ 0 \leq x_2 \leq N_{\text{ports}} \cdot C_{\text{single}}, \\ T_{\text{min}} = \frac{M_{\text{goal}}}{L_{\text{max}} + N_{\text{ports}} \cdot C_{\text{single}}}, \\ T_{\text{max}} = \frac{M_{\text{goal}}}{N_{\text{ports}} \cdot C_{\text{single}}}, \\ x_1, x_2 \in \mathbb{R}^+, w_1 + w_2 = 1. \end{cases} \end{aligned} \quad (12)$$

where x_1 and x_2 are the decision variables representing the annual mass transport rates for rockets and elevators, respectively. w_1 and w_2 are weighting parameters.

The formulation components are:

1. $\mathcal{C}(x_1, x_2)$ is the normalized cost, defined as the average unit cost of the mixed strategy relative to the rocket-only unit cost. Since $C_2 \ll c_1$, increasing elevator flow x_2 drives $\mathcal{C}$ toward its lower bound C_2/c_1 , reflecting the elevator's advantage.
2. $\mathcal{T}(x_1, x_2)$ is the normalized time, proportional to the inverse of total throughput $(x_1 + x_2)$. It penalizes low flux by increasing Z as transport slows.
3. $w_1, w_2 > 0$ with $w_1 + w_2 = 1$ control the cost-time trade-off. Larger w_2 prioritizes faster completion over savings.
4. Constraints enforce feasibility: $x_2 \leq N_{\text{ports}} C_{\text{single}}$ limits Galactic Harbour capacity, and $x_1 \leq L_{\text{max}}$ enforces the global launch cap.

4.2 Analysis of Transport Scenarios

We apply the proposed model to the three required scenarios. To solve this non-convex problem while avoiding local optima, we use Differential Evolution as a robust global optimizer. The solver uses an adaptive population size based on variable bounds, with a maximum of 1000 iterations and a convergence tolerance of 10^{-10} .

We solve the non-convex ILCM using a two-stage hybrid routine: Differential Evolution [7] for global exploration followed by L-BFGS-B for local refinement. Algorithm 1 outlines the solver used for Scenarios A–C.

Algorithm 1: Hybrid Optimization Algorithm for ILCM Solver

Input: Target Mass M_{goal} , Weights w_1, w_2 , Cost Functions $c_1(t), C_2(t)$, Capacity Bounds

Output: Optimal Transport Rates $x^* = [x_1^*, x_2^*]$

- 1 **Step 1: Global Optimization (Differential Evolution);**
- 2 Initialize population P with $N = 15$ individuals within bounds $[0, L_{max}]$;
- 3 Set parameters: Strategy='best1bin', $MaxIter = 1000$, $Tol = 1e - 10$;
- 4 **while** termination criteria not met **do**
- 5 **for** each candidate solution $\mathbf{x}_i \in P$ **do**
- 6 Calculate Project Duration: $T_{end} = 2050 + M_{goal}/(x_{1,i} + x_{2,i})$;
- 7 Compute Definite Integrals of cost functions: $I_k = \int_{2050}^{T_{end}} c_k(t)dt$ for $k = 1, 2$;
- 8 Evaluate Fitness $Z(\mathbf{x}_i) \leftarrow w_1 \cdot C_{norm}(\mathbf{x}_i) + w_2 \cdot T_{norm}(\mathbf{x}_i)$;
- 9 **end**
- 10 Perform Mutation and Crossover to generate offspring;
- 11 Select best individuals for next generation;
- 12 **end**
- 13 $\mathbf{x}_{DE} \leftarrow$ Best individual from DE;
- 14 **Step 2: Local Refinement (L-BFGS-B);**
- 15 Using $\mathbf{x}_{DE}$ as initial guess, execute gradient-based minimization;
- 16 Refine solution to satisfy constraints with precision $1e - 12$;
- 17 **return** $\mathbf{x}^*$;

4.2.1 Scenario A: Space Elevator System Only

In this scenario, we strictly set $x_1 = 0$. The transport relies entirely on the three Galactic Harbours. Therefore, Eq. (12) can be reduced to the following form:

$$\min_{x_2} \quad Z = w_1 \underbrace{\left(\frac{\int_{2050}^{\frac{M_{goal}}{x_2} + 2050} C_2(t)x_2 dt}{L_{max} \int_{2050}^{T_{min} + 2050} c_1(t) dt} \right)}_{C(0, x_2)} + w_2 \underbrace{\left(\frac{M_{goal}}{T_{max} x_2} \right)}_{T(0, x_2)} \quad (13)$$

$$\text{s.t.} \quad \begin{cases} 0 \leq x_2 \leq N_{ports} \cdot C_{single}, \\ x_2 \in \mathbb{R}^+, \\ w_1 + w_2 = 1. \end{cases}$$

4.2.2 Scenario B: Traditional Rockets Only

In this scenario, the Space Elevator is unavailable ($x_2 = 0$). Transport relies on the 10 global launch sites, Eq. (12) can be reduced to:

$$\begin{aligned} \min_{x_1} \quad Z &= w_1 \underbrace{\left(\frac{\int_{2050}^{\frac{M_{\text{goal}}}{x_1} + 2050} c_1(t) x_1 \, dt}{L_{\text{max}} \int_{2050}^{T_{\text{min}} + 2050} c_1(t) \, dt} \right)}_{\mathcal{C}(x_1, 0)} + w_2 \underbrace{\left(\frac{M_{\text{goal}}}{T_{\text{max}} x_1} \right)}_{\mathcal{T}(x_1, 0)} \\ \text{s.t.} \quad &\begin{cases} 0 \leq x_1 \leq L_{\text{max}}, \\ x_1 \in \mathbb{R}^+, \\ w_1 + w_2 = 1. \end{cases} \end{aligned} \quad (14)$$

4.2.3 Scenario C: Combined Strategy

This scenario allows for the simultaneous optimization of x_1 and x_2 , The constrained problem is given in Eq. (12):

$$\begin{aligned} \min_{x_1, x_2} \quad Z &= w_1 \underbrace{\left(\frac{\int_{2050}^{\frac{M_{\text{goal}}}{x_1 + x_2} + 2050} c_1(t) x_1 \, dt + \int_{2050}^{\frac{M_{\text{goal}}}{x_1 + x_2} + 2050} C_2(t) x_2 \, dt}{L_{\text{max}} \int_{2050}^{T_{\text{min}} + 2050} c_1(t) \, dt} \right)}_{\mathcal{C}(x_1, x_2)} + w_2 \underbrace{\left(\frac{M_{\text{goal}}}{T_{\text{max}}(x_1 + x_2)} \right)}_{\mathcal{T}(x_1, x_2)} \\ \text{s.t.} \quad &\begin{cases} 0 \leq x_1 \leq L_{\text{max}}, \\ 0 \leq x_2 \leq N_{\text{ports}} \cdot C_{\text{single}}, \\ x_1, x_2 \in \mathbb{R}^+, \\ w_1 + w_2 = 1. \end{cases} \end{aligned} \quad (15)$$

Table 3: Comparative Performance Metrics of Three Transport Scenarios

Scenario	x_1 (kt/year)	x_2 (kt/year)	Duration (years)	Completion (Year)	Total Cost (USD)	Unit Cost (\$/tonne)
A: Elevator Only	0	537.0	186.22	2236	7.24×10^{11}	7.24×10^3
B: Rockets Only	979.8	0	102.06	2152	8.59×10^{13}	8.59×10^5
C: Combined	439.2	537.0	102.43	2152	3.89×10^{13}	3.89×10^5
Relative Performance (vs. Scenario C)						
Scenario A	-100%	0%	+81.8%	+84 yrs	-98.1%	-98.1%
Scenario B	+123.2%	-100%	-0.4%	~ 0 yrs	+120.8%	+120.8%

Key Insights from Comparative Analysis: Table 3 and Figure 6 reveal a distinct cost-time Pareto frontier. Scenario A achieves the minimum cost ($\$7.24 \times 10^{11}$) but suffers an 82% increase in timeline. Scenario B minimizes the duration (102 years) but at a 119-fold increase in cost. Scenario C strikes the best balance, reducing costs by 55% compared to B while maintaining the same speed. This demonstrates that a hybrid strategy, which leverages the Space Elevator for efficient baseload transport and uses rockets for critical, time-sensitive deliveries, offers an optimal solution.

Scenario C strikes the best balance, reducing costs by 55% compared to B while maintaining the same speed. This demonstrates that a hybrid strategy, which leverages the Space Elevator for efficient baseload transport and uses rockets for critical, time-sensitive deliveries, offers an optimal solution.

The trade-off highlights the importance of combining the strengths of both transport methods to meet both cost and time objectives. Additionally, it suggests that relying solely on rockets would result in unsustainable costs, while relying solely on the Space Elevator would lead to significantly longer construction times.

This reinforces the need for a balanced, multi-modal approach in large-scale interplanetary logistics, ensuring that both financial and temporal constraints are effectively managed.

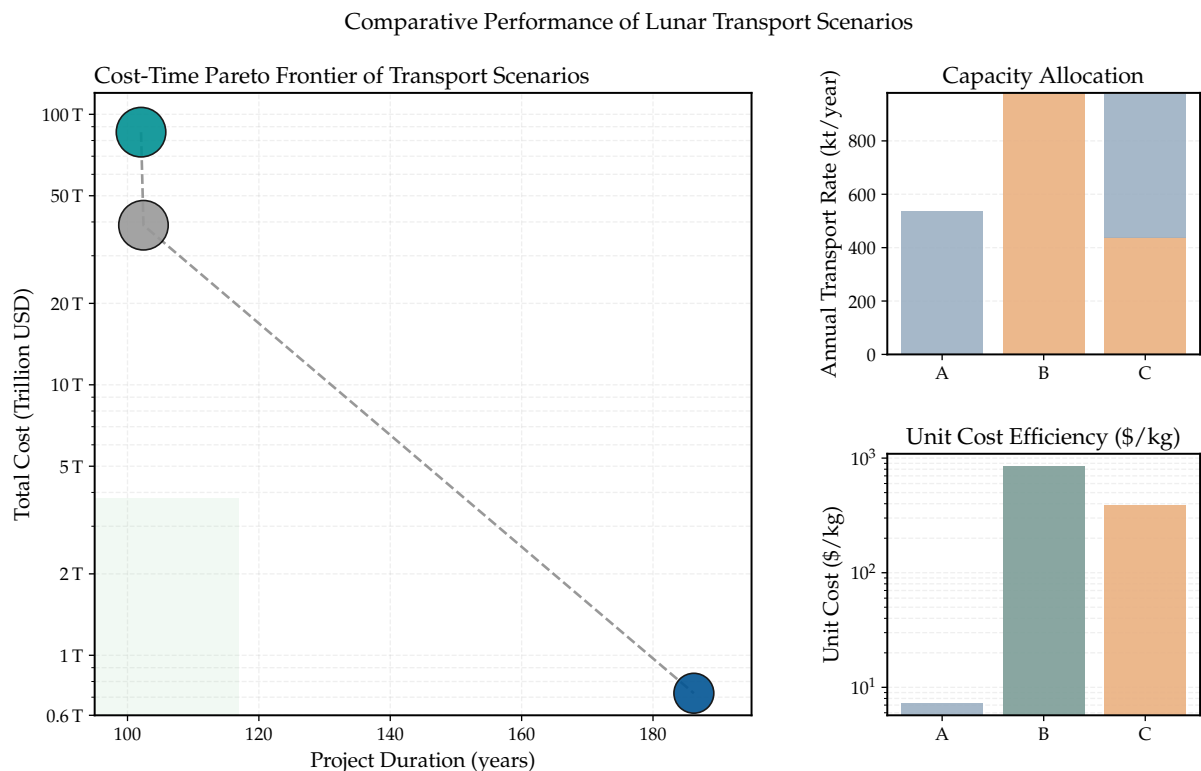


Figure 6: Comparative Analysis of Transport Scenarios: Cost vs. Duration Trade-offs. The plot highlights Scenario C as the Pareto-optimal compromise.

5 Task 2: Analysis of Non-Ideal Operational Scenarios

To address system performance under non-ideal conditions, we transition from a deterministic baseline to a stochastic framework. By relaxing the assumption of perfect reliability, we explicitly model critical disruptions, including launch failures, tether swaying, and mechanical breakdowns, as probabilistic events. This analysis aims to quantify how these uncertainties alter the logistics strategy, project timeline, and overall cost relative to the ideal scenario. In doing so, the framework captures variability

in system outcomes that cannot be represented by deterministic models. The results provide a more realistic basis for risk-aware planning and decision-making.

5.1 Stochastic Modeling of Operational Disturbances

We characterize the non-ideal operational states by defining specific efficiency coefficients, where α represents the performance of traditional rockets and β corresponds to the Space Elevator. These parameters serve to quantify the effective throughput capacity under uncertainty.

5.1.1 Quantification of Non-Ideal Model Corrections

To transition from a theoretical baseline to a robust operational framework, we introduce four parallel model corrections. These corrections quantify the physical throughput constraints and economic penalties driven by system stochasticity.

Correction I: Mechanical Availability Constraints (η_{mech}). Drawing parallels to high-reliability terrestrial continuous transport systems (e.g., maglevs), we adopt a steady-state mechanical availability of $\eta_{\text{mech}} = 0.98$. This parameter accounts for routine climber maintenance, handling delays at the Earth Port, and stochastic component failures, assuming a mature operational phase with redundant climber fleets [8].

Correction II: Environmental Interruption Constraints (A_{tether}). Pursuant to Edwards' design principles [9], system interruptions are necessitated only by trackable large debris (> 10 cm) and severe weather [10]. We model these mandatory pauses as a Poisson process with an annual frequency $\lambda = 36.5$ and a mean downtime $\mu = 0.083$ days. Integrating this with Correction I, the comprehensive availability coefficient β

$$A_{\text{tether}} \approx 1 - \frac{\lambda\mu}{365} \approx 0.9917, \quad \beta = \eta_{\text{mech}} A_{\text{tether}} \approx 0.972.$$

Hence the effective elevator-capacity constraint becomes

$$x_2 \leq \beta \cdot N_{\text{ports}} \cdot C_{\text{single}} \quad (16)$$

Correction III: Reliability-Adjusted Rocket Cost ($\hat{C}_1$). Under the Expected Cost Model, the financial burden of a successful delivery must internalize the probabilistic cost of failure. Consequently, the effective unit cost of rocket transport is inflated inversely to its success reliability α :

$$\hat{C}_1(t) = \frac{C_1(t)}{\alpha} \quad (17)$$

Correction IV: Availability-Adjusted Elevator Cost ($\hat{C}_2$). Leveraging the LCOT methodology [11], we decouple the Space Elevator's cost structure based on throughput sensitivity. Variable energy costs (c_{energy}) remain invariant, while inelastic annual expen-

ditures (Capital, O&M, Risk) are penalized by the reduced utilization factor β . The robust unit cost function is formulated as:

$$\hat{C}_2(t) = C_{\text{energy}}(t) + \frac{1}{\beta} [C_{\text{O\&M}}(t) + C_{\text{cap}}(t) + C_{\text{risk}}(t)] \quad (18)$$

5.2 Formulation of the Robust Optimization Model

Incorporating the derived disturbance coefficients, we reformulate the baseline model into a robust framework that captures the dual impact of non-ideal conditions: capacity constriction and cost escalation. Specifically, the effective throughput upper bounds are reduced by the efficiency factors (α, β) , while the effective unit costs are inversely adjusted to $\hat{C}_1(t)$ and $\hat{C}_2(t)$ to reflect operational expenditures incurred during failure states. The modified robust optimization problem is formulated as follows:

$$\begin{aligned} \min_{x_1, x_2} \quad & Z = w_1 \underbrace{\left(\frac{\int_{2050}^{\frac{M_{\text{goal}}}{x_1+x_2}+2050} \hat{C}_1(t)x_1 dt + \int_{\frac{M_{\text{goal}}}{x_1+x_2}+2050}^{2050} \hat{C}_2(t)x_2 dt}{\alpha \cdot L_{\text{max}} \int_{2050}^{T_{\text{min}}+2050} \hat{C}_1(t) dt} \right)}_{\mathcal{C}(x_1, x_2)} + w_2 \underbrace{\left(\frac{M_{\text{goal}}}{T_{\text{max}}(x_1 + x_2)} \right)}_{\mathcal{T}(x_1, x_2)} \\ \text{s.t.} \quad & \begin{cases} 0 \leq x_1 \leq \alpha \cdot L_{\text{max}}, \\ 0 \leq x_2 \leq \beta \cdot N_{\text{ports}} \cdot C_{\text{single}}, \\ x_1, x_2 \in \mathbb{R}^+, \\ w_1 + w_2 = 1, \\ T_{\text{min}} = \frac{M_{\text{goal}}}{\alpha \cdot L_{\text{max}} + \beta \cdot N_{\text{ports}} \cdot C_{\text{single}}}, \\ T_{\text{max}} = \frac{M_{\text{goal}}}{\beta \cdot N_{\text{ports}} \cdot C_{\text{single}}}. \end{cases} \end{aligned} \quad (19)$$

Solving the robust model (Eq. (19)) with efficiency coefficients $\alpha = 0.9757$ and $\beta = 0.972$ yields adjusted transport rates of $x_1 = 431,303$ and $x_2 = 521,964$ tons/year. The project duration extends to 104.90 years (completion ≈ 2155)—a slight delay compared to the 102.43-year baseline—while total costs rise to $\$3.92 \times 10^{13}$.

5.2.1 Comparative Analysis

Relative to the ideal baseline (Scenario C), realistic disturbances induce a 2.4-year delay and a marginal cost increase ($\approx 0.8\%$). Despite capacity reductions ($\alpha, \beta < 1$) and cost penalties, the strategy proves highly resilient. Reliability losses are mitigated by extending the timeline rather than drastically altering the transport mix: rocket usage decreases slightly ($\sim 439\text{k} \rightarrow \sim 431\text{k}$) while elevator usage adjusts ($537\text{k} \rightarrow \sim 522\text{k}$). This confirms the ILCM framework's robustness against nominal operational friction.

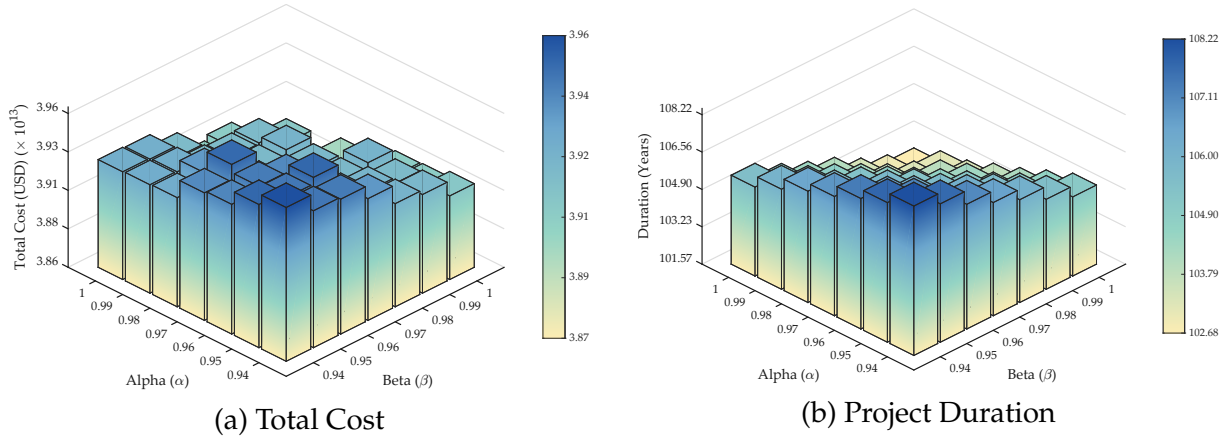


Figure 7: Sensitivity Analysis Results. (a) Total Cost variation under different α, β ; (b) Project Duration variation under different α, β .

5.2.2 Sensitivity Landscape Analysis

Figure 7 illustrates the sensitivity of Total Cost and Project Duration to reliability parameters α and β . The analysis confirms a distinct monotonic relationship: as component reliability decreases, both metrics exhibit a stable, predictable increase. The step-wise degradation suggests manageable system behavior. Specifically, Figure 7(a) indicates that while financial penalties accumulate due to increased reliance on backup rockets, the overall budget remains within the same order of magnitude even under worst-case scenarios ($\alpha, \beta \approx 0.94$). Similarly, Figure 7(b) reflects a proportional reduction in effective transport capacity, thereby validating the linear robustness of the hybrid logistics network.

6 Task 3: Scenario Analysis of Water Sustainability

To account for the "additional cost and timeline" of water requirements, we model the Moon Colony's hydrological system as a dynamic metabolic [12] process, distinct from the static logistics of construction materials. The temporal baseline is set at full structural completion. Our analysis divides the water strategy into two phases: the pre-operational reserve and long-term metabolic evolution.

6.1 Phase I: Pre-Habitation Strategic Buffer Deployment

Before the colony initiates operations with 100,000 initial residents, a hydrological safety margin must be pre-positioned to mitigate supply chain risks.

6.1.1 Buffer Sizing Logic

We define the strategic reserve V_{buffer} as 90 days of standard full-capacity consumption. This horizon follows NASA's Human Integration Design Handbook guidance for deep-space logistics resilience, bridging typical Earth–Moon transport windows and providing redundancy against Space Elevator downtime. Using $W_{std} = 25$ L/person/day [13], the static mass is:

$$V_{buffer} = P_{initial} \cdot W_{std} \cdot T_{safety} = 100,000 \times 0.025 \times 90 = 225,000 \text{ tons} \quad (20)$$

6.1.2 Transport Cost and Timeline

Treating the strategic water reserve V_{buffer} as an added payload in the queue, we apply the robust logistics framework from Task 2. Using Year 2155 parameters with Space Elevator efficiency $\beta \approx 0.972$, we obtain:

- **Additional Cost:** Substituting the mass into $C_2(t)$ gives

$$\text{Cost}_{\text{water}} = 225,000 \text{ tons} \times \$7013/\text{ton} \approx \$1.578 \text{ Billion.}$$

- **Additional Time:** With adjusted throughput $x_2(t)$,

$$T_{\text{water}} = \frac{225,000}{522,000} \approx 0.43 \text{ years.}$$

This transport is scheduled concurrently with habitat interior outfitting, so it causes zero delay on the critical path.

6.2 Phase II: Long-term Dynamic Metabolic Evolution

After construction, the colony transitions to a dynamic equilibrium. To quantify long-term resource dependence, we introduce the Phased Dynamic Evolution Framework to model the annual external resupply requirement $R(t)$.

Figure 8 illustrates the physical basis: a semi-closed loop coupling biological metabolism, wastewater reclamation, and ISRU inputs. The performance of these subsystems determines the macroscopic recycling parameter $\eta(t)$ used in our formulation.

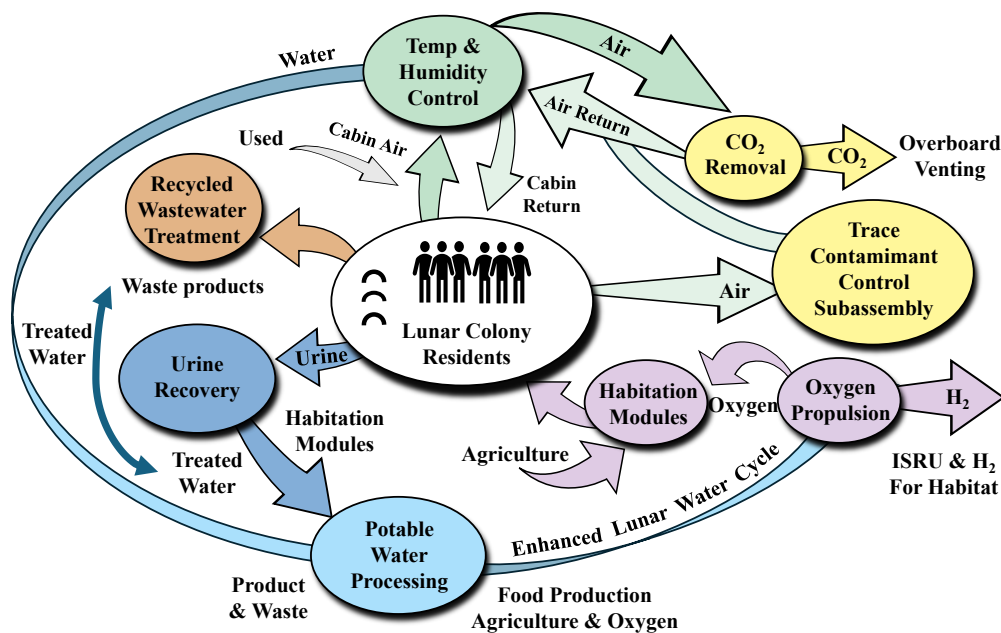


Figure 8: Architecture of the Enhanced Lunar Water Cycle.

6.2.1 Mathematical Formulation

1. Population Growth We use a logistic model with initial population $P_0 = 100,000$ and carrying capacity $K = 1,000,000$:

$$N(t) = \frac{K}{1 + \left(\frac{K-P_0}{P_0}\right) e^{-0.08t}} \quad (21)$$

2. Accelerated Demand Evolution Per capita demand $W(t)$ evolves according to Bass diffusion and resource abundance effects:

$$W(t) = \begin{cases} 20 & 0 \leq t < 3 \\ 20 + 8.89(t - 3) & 3 \leq t < 12 \\ 100 + 150 \sin\left(\frac{\pi(t-12)}{36}\right) & 12 \leq t < 30 \\ 250 & t \geq 30 \end{cases} \quad [\text{L/person/day}] \quad (22)$$

This achieves Earth-normal consumption by Year 30, influenced by governance ($p = 0.05$, $q = 0.7$) and ISRU supply elasticity (10^{11} kg reserves).

3. System Efficiency & Resupply Recycling efficiency $\eta(t)$ approaches its theoretical limit, defining the net system deficit $R(t)$:

$$\begin{aligned} \eta(t) &= 0.999 - 0.149 \cdot e^{-0.08t} \\ R(t) &= N(t) \cdot W(t) \cdot [1 - \eta(t)] \cdot 0.365 \quad [\text{tons/year}] \end{aligned}$$

6.2.2 Simulation Results

Figure 9 visualizes the system dynamics over 70 years, highlighting the Amplified “Resupply Hump.” The left panel shows how the interplay between logistic population growth ($N(t)$, blue dashed), accelerated demand ($W(t)$, teal), and recycling efficiency ($\eta(t)$, purple solid) leads to a non-linear resupply requirement ($R(t)$), represented by the red region.

The bottom right bar chart identifies a 24-year Logistical Stress Window (Years 14–37), where annual demand exceeds infrastructure capacity (orange dashed line, 537 kt/yr), reaching a maximum deficit of 826.5 kt in Year 24. This surge necessitates supplementary rocket deployment, as outlined in our robust model, to bridge the gap until efficiency saturation.

Table 4: Key Milestones and the Identification of Peak Stress Window

Year	Phase	$N(t)$ (k)	$W(t)$ (L/d)	$\eta(t)$ (%)	$R(t)$ (kt/yr)
0	I. Survival	100	20	85.0	109.5
12	II. Ramp-up	276	100	93.8	625.2
24	III. Peak Stress	431	230	97.7	826.5
30	III. Transition	551	250	98.6	788.8
50	IV. Mature	864	250	99.4	473.7
70	IV. Stable	968	250	99.7	265.2

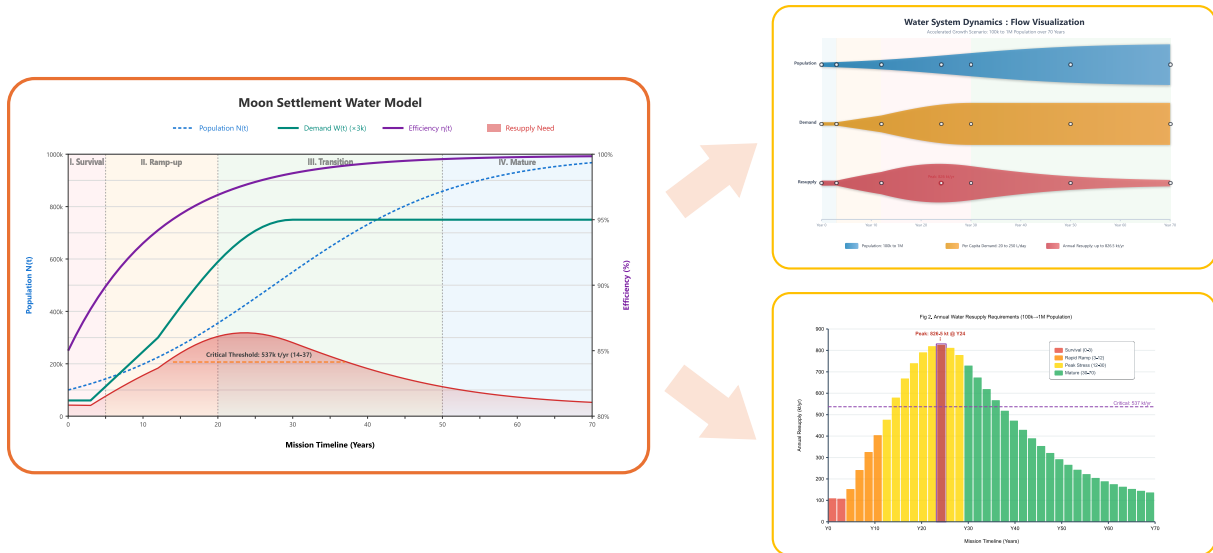


Figure 9: Integrated Water System Dynamics and Capacity Deficit Analysis.

This observed peak results from the Scissors Effect, where a 150% demand surge outpaces efficiency gains, causing a deficit that exceeds baseline capacity by 62%. However, by Year 40, the system achieves a stable resource-population decoupling. As recycling efficiency gradually saturates at $\approx 99.7\%$, resupply needs drop to 265 kt/yr, confirming the colony's long-term sustainability.

6.2.3 Transport Strategy and Capacity Analysis

The deficit during the "Stress Window" requires a hybrid transport architecture:

$$\text{Peak Utilization Ratio} = \frac{R(24)}{C_{\text{elevator}}} = \frac{826,500}{522,000} \approx 158.3\%$$

6.2.4 Hybrid Transport Allocation Strategy

When demand exceeds the Space Elevator's capacity by 58.3%, a phased allocation strategy is proposed. In the Pre-Critical Phase (Years 0–13), transport relies solely on the Space Elevator at full capacity, with no rockets. During the Critical Window (Years 14–37), the Space Elevator operates at maximum capacity (522 kt/year), supplemented by rockets. In the Post-Critical Phase (Years 38+), the system returns to full Space Elevator capacity with no rockets.

6.2.5 Cost Function by Phase

The additional cost to meet water demand is calculated based on the resupply rate $R(t)$. In the Pre-Critical Phase (Years 0–13), cost is proportional to $R(t) \cdot C_2(t + 105)$. During the Critical Window (Years 14–37), the cost combines Space Elevator and rocket costs, with $C_2(t + 105) \cdot \alpha \cdot N_{\text{ports}} \cdot C_{\text{single}}$ and $c_1(t + 105)$ for rocket demand. In the Post-Critical Phase (Years 38+), the cost remains proportional to $R(t) \cdot C_2(t + 105)$.

7 Task4: Environmental Impact Analysis

The baseline ILCM prioritizes efficiency, neglecting the ecological impacts of heavy-lift launches like black carbon and alumina injection. We introduce an environmental objective $\mathcal{E}$, creating a multi-objective trade-off between colonization speed and biosphere preservation. This analysis quantifies the "sustainability cost" and determines the threshold for zero-emission logistics.

7.1 Model Adjustment for Sustainability

The non-ideal ILCM model is adjusted by introducing a third objective term, $\mathcal{E}(x_1, x_2)$, representing the Environmental Impact, which is defined as the ratio of mass transported by conventional rockets (high-polluting) to the clean Space Elevator.

The modified objective function is:

$$\begin{aligned} \min_{x_1, x_2} \quad Z = & w_1 \underbrace{\left(\frac{\int_{2050}^{\frac{M_{\text{goal}}}{x_1+x_2}+2050} \hat{C}_1(t)x_1 dt + \int_{2050}^{\frac{M_{\text{goal}}}{x_1+x_2}+2050} \hat{C}_2(t)x_2 dt}{\alpha \cdot L_{\text{max}} \int_{2050}^{T_{\text{min}}+2050} \hat{C}_1(t) dt} \right)}_{\mathcal{C}(x_1, x_2)} \\ & + w_2 \underbrace{\left(\frac{M_{\text{goal}}}{T_{\text{max}}(x_1 + x_2)} \right)}_{\mathcal{T}(x_1, x_2)} + w_3 \underbrace{\left(\frac{x_1}{x_1 + x_2} \right)}_{\mathcal{E}(x_1, x_2)} \quad (23) \\ \text{s.t.} \quad & \begin{cases} 0 \leq x_1 \leq \alpha \cdot L_{\text{max}}, & 0 \leq x_2 \leq \beta \cdot N_{\text{ports}} \cdot C_{\text{single}}, \\ x_1, x_2 \in \mathbb{R}^+, & w_1 + w_2 + w_3 = 1, \\ T_{\text{min}} = \frac{M_{\text{goal}}}{\alpha \cdot L_{\text{max}} + \beta \cdot N_{\text{ports}} \cdot C_{\text{single}}}, & T_{\text{max}} = \frac{M_{\text{goal}}}{\beta \cdot N_{\text{ports}} \cdot C_{\text{single}}}. \end{cases} \end{aligned}$$

where x_1 represents rocket payload (tons/year) and x_2 represents Space Elevator payload. The weight w_3 serves as a tunable policy parameter to represent the priority of environmental protection.

7.2 Scenario Analysis: The Trade-off

We performed a sensitivity analysis on the environmental weight w_3 (with $w_1 = w_2$) to simulate different policy scenarios. The results in Figure 10 show a clear trade-off between environmental impact and transport capacity.

- **Low Standard** ($w_3 < 0.1$): The model prioritizes speed and capacity, using rockets near their maximum ($\sim 400,000$ t/year). This achieves the fastest colony goal but causes the highest atmospheric damage.
- **Transition Zone** ($0.1 < w_3 < 0.26$): As environmental penalties rise, the optimal strategy shifts. Rocket usage decreases linearly, reflecting a "mixed fleet" approach where rockets are used only for time-critical payloads.
- **Zero-Emission Threshold** ($w_3 > 0.262$): At $w_3 \approx 0.262$, rocket transport is fully abandoned ($x_1 \rightarrow 0$), relying solely on the Space Elevator. This minimizes environmental impact but increases project duration by 29%.

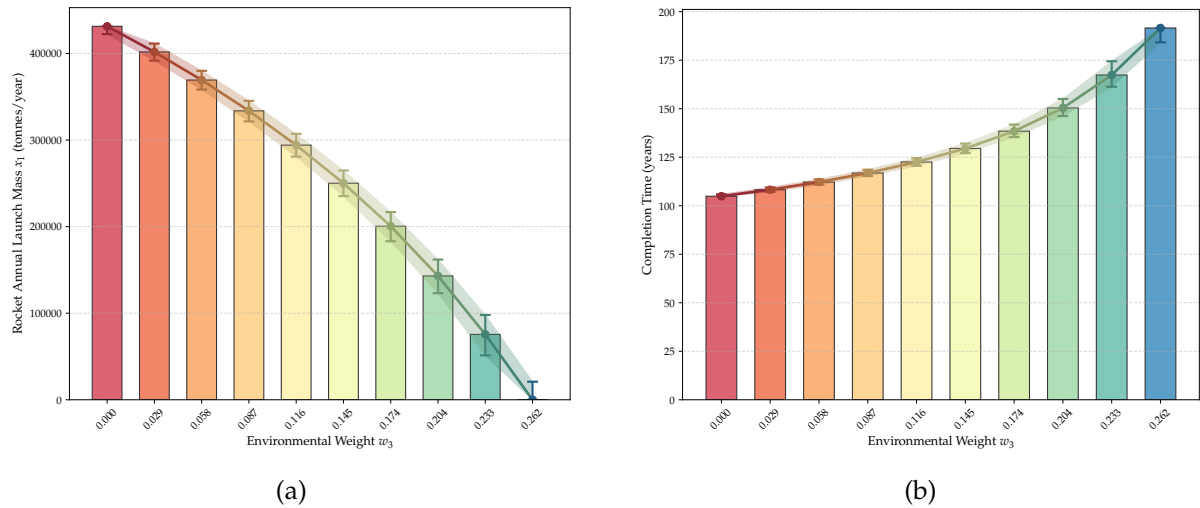


Figure 10: Sensitivity Analysis of Environmental Policy. (a) Decline in high-polluting rocket usage as w_3 increases, reaching zero at $w_3 \approx 0.262$. (b) The trade-off in extended project duration due to reliance on the slower Space Elevator.

7.3 Minimizing Environmental Impact

To minimize environmental impact, our model suggests weighting pollution cost (w_3) at least 26.2% in decision-making. Without significant environmental penalties (e.g., carbon taxes or strict quotas), economic and temporal pressures favor high-pollution rocket usage. The Space Elevator is crucial for a sustainable colony, but its lower throughput requires a longer timeline to meet sustainability goals.

8 Conclusion

This paper presents the ILCM to optimize the transport of 10^8 metric tons for a self-sustaining lunar civilization. The analysis shows that the choice between rockets and space elevators is not binary but a trade-off that requires a hybrid approach. While Space Elevators offer a cost-effective solution for bulk cargo, their slow deployment makes them unsuitable for early colonization phases. In contrast, relying solely on rockets is financially unsustainable. The proposed Hybrid Strategy effectively combines the strengths of both, using Space Elevators for heavy-lift baseloads and rockets to compress the construction timeline by 84 years compared to an elevator-only approach.

The model is robust against disruptions like launch failures and tether downtimes, with a dual-channel architecture proving resilient. A critical "Water Stress Window" was identified, where the rocket fleet bridges the gap until the recycling system matures. Incorporating environmental objectives, the model emphasizes shifting to solar-powered elevators to minimize pollution. In conclusion, the ILCM framework demonstrates that a 100,000-person lunar colony is both technically feasible and economically viable through a balanced, resilient, and multi-modal logistics system.

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MEMO

TO: MCM Agency Executive Board

FROM: Modeling Team # 2601969

DATE: February 2, 2026

SUBJECT: Strategic Roadmap for Lunar Colony Logistics

Based on our **ILCM analysis**, single-mode strategies for the 100,000-person Moon Colony are unsustainable. We propose a **Synergistic Hybrid Strategy**—annually transporting 537,000 tonnes via Space Elevator and 439,200 tonnes via rockets—which completes construction in 102 years at a cost of **\$38.9 trillion**, achieving a **55% cost reduction** and an **84-year timeline compression**.

Beyond core logistics, this dual-channel architecture ensures critical **operational resilience**. Maintaining active rocket production lines safeguards supply continuity during Space Elevator maintenance or disruptions. This redundancy is vital to navigate the identified **Water Stress Window**—a 24-year period where demand temporarily outpaces recycling capacity, creating an 826,500-tonne resource gap. The rocket fleet provides the necessary surge capacity to bridge this gap, ensuring colony stability during its most vulnerable expansion phase.

From an **environmental stewardship** perspective, shifting bulk transport to the solar-powered Space Elevator drastically reduces stratospheric pollution from rocket launches. Our modeling indicates that prioritizing emissions reduction by assigning it a **26.2%** weight in our objective function would eliminate rocket use entirely, though it would extend the project timeline by 29%. This highlights the essential role of the hybrid strategy in harmonizing project ambitions with planetary sustainability.

Critical Implementation Steps include the immediate authorization of foundational construction for the **Galactic Harbour**, strategic procurement of flexible rocket launch contracts to support both **baseline and surge operations**, establishment of a dedicated oversight committee to continuously optimize the **transport allocation ratio**, and development of **contingency protocols** for non-ideal operational scenarios. This resilient, multi-modal logistics framework not only represents a technically feasible solution but also provides a **sustainable pathway** that harmonizes lunar settlement ambitions with fiscal prudence and environmental responsibility.

This resilient, multi-modal logistics framework represents not only a technically feasible solution but also a sustainable pathway that harmonizes lunar settlement ambitions with fiscal prudence and environmental responsibility.

Respectfully submitted,

Modeling Team # 2601969

Lunar Colonization Optimization Task Force

A Use of AI in Model Development and Optimization

This section documents the interactions between the author and AI assistant during the development of this MCM/ICM paper, highlighting the role of AI tools in literature review, model optimization, and language refinement.

AI Tool Used

Question: Which AI tool did you use?

Which AI tool did you use?

AI Response

We used ChatGPT-5.2 provided by OpenAI.

Purpose of Using AI

Question: What was the purpose of using this AI tool?

What was the purpose of using this AI tool?

AI Response

- Literature Search: To quickly review and understand the background of aerospace engineering and related technologies, such as Space Elevators and rocket propulsion for lunar transport. - Code Assistance: To help debug and optimize the Python script for the ILCM algorithm, particularly for implementing Differential Evolution. - Language Polishing: To improve the academic tone and grammatical accuracy of sections like the Abstract and Conclusion, ensuring clarity and conciseness.

Specific Prompts Used

Question: What were the specific prompts used?

What were the specific prompts used?

AI Response

- Prompt 1 (Background): "Survey the current literature on Space Elevator concepts and their projected economic advantages over traditional chemical rockets for lunar transport." - Prompt 2 (Coding): "Optimize the Python code implementation of the Differential Evolution algorithm for efficiency and performance."

- Prompt 3 (Refining): "Rewrite the description of the 'Independent Logistics Channel Model (ILCM)' to be more concise and suitable for a mathematical modeling competition paper."

Evaluation and Modification of AI Output

Question: How did you evaluate and modify the AI's output?

How did you evaluate and modify the AI's output?

AI Response

All information provided by the AI was cross-checked against peer-reviewed journals and authoritative sources in the field. For the coding assistance, we manually tested and integrated the AI-generated code into our model framework, ensuring compatibility and accuracy. The final version of the paper, including all logic and conclusions, was reviewed and refined by all team members to guarantee its originality and validity.